

Worksheet — The Root System

Student Name: _____ Date: _____

Section: _____ Class/Roll No: _____

A. Fill in the Blanks

1. The primary root develops from the embryonic _____.
2. Fibrous roots are characteristic of _____ (dicots/monocots).
3. The region of the root tip that protects the growing apex is the _____.
4. Root hairs are found in the region of _____.

B. One Word Answer

1. Name the root system found in mustard. Ans: _____
2. Name the root system found in wheat. Ans: _____
3. Name the roots that arise from stem/leaf tissue instead of the radicle. Ans: _____
4. Name the structure that covers the root apex. Ans: _____

C. Match the Following

Column A	Column B
Tap root	Wheat
Fibrous root	Mustard
Adventitious root	Banyan
Root hair	Absorption

D. True or False

1. Root hairs develop in the region of meristematic activity. (True/False)
2. The tap root system is typical of monocotyledonous plants. (True/False)
3. Adventitious roots always arise from the radicle. (True/False)
4. The root cap protects the growing root tip. (True/False)

E. Short Answer Questions

1. Differentiate between tap root and fibrous root systems. [2 Marks]
2. What are adventitious roots? Give two examples. [2 Marks]

F. Short Essay Questions

1. Describe the four structural regions of a root tip. [3 Marks]
2. Explain any three functions of the root system with examples. [3 Marks]

G. Long Answer / Essay Questions

1. With the help of a labeled diagram, describe the structure of a dicot root tip and explain the significance of each region. [5 Marks]